

Figitumumab combined with carboplatin and paclitaxel in treatment-naïve Japanese patients with advanced non-small cell lung cancer

Summary Objectives The insulin-like growth factor (IGF) signaling pathway has been implicated in the pathogenesis of numerous tumor types, including non-small cell lung cancer (NSCLC). Figitumumab is a fully human IgG2 monoclonal antibody against IGF-1 receptor (IGF-1R). **Methods** This phase I, open-label, dose-escalation study (ClinicalTrials.gov: NCT00603538) assessed the safety and tolerability of figitumumab (6, 10 and 20 mg/kg) in combination with carboplatin (area under the curve: 6 mg·min/mL) and paclitaxel (200 mg/m²) in Japanese patients (N = 19) with chemotherapy-naïve, advanced NSCLC. Treatments were administered intravenously on day 1 of a 21-day cycle for four to six cycles. Pharmacokinetics, biomarkers, and antitumor activity were also evaluated. **Results** Figitumumab in combination with carboplatin and paclitaxel was well tolerated at doses up to 20 mg/kg; no dose-limiting toxicities were observed at this dose level. When given in combination, figitumumab plasma exposure increased in an approximately dose-proportional manner. The approximate 2-fold accumulation following repeated administration supported the 21-day regimen as appropriate for figitumumab administration. Serum total IGF-1 and IGF binding protein-3 concentrations increased following figitumumab dosing, but a clear dose-dependent relationship was not demonstrated. Seven of 18 evaluable patients experienced a partial response. **Conclusions** Figitumumab 20 mg/kg in combination with carboplatin and paclitaxel was well tolerated in chemotherapy-naïve Japanese patients with NSCLC. Further analysis of biomarker data is necessary for the development of figitumumab therapy.